K-Fire Hydraulic Solver

Hydraulic Calculation Engineering Overview

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Revision History

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1.0.0	2026-07-10	4.4.3	First edition of Engineering Overview for distribution attachment
1.0.1	2026-07-10	4.4.3	Adjusted disclosure level and length; packaged figures and Word document
1.1.0	2026-07-11	4.4.4	Added dedicated Antifreeze analysis (§6.6)
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1.3.0	2026-10-23	5.8.0	Removed the design-area sweep on grids. Automatic sizing uses only the design area that is currently active, and branch lines and cross mains on a grid are each unified to the largest diameter in the group.
1.4.0	2026-10-26	6.0.0	Added pipe geometry

			editing (§3.5: 45° diagonal drawing, rotation of pipe runs, risers, and tees, 30-step undo) and report output (§8.5: whole-network drawing on the cover, notice for scenarios without results).
1.5.0	2026-10-29	6.2.0	Added relief valve (§3.6) and flow control valve (§3.7).

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0. Introduction

0.1 Document purpose and scope of application

This document explains on what engineering principles the steady-state hydraulic calculation function of K-Fire Hydraulic Solver is constructed. While the user manual covers screen operations, this material provides the physical basis and scope of interpretation needed when reviewing calculation results or explaining them to third parties.

Steady-state hydraulic calculation determines the distribution of flow and pressure at a stable operating point in a water-filled piping network. Its purpose differs from functions that address time-dependent filling and air discharge, such as Water Delivery Time (WDT) for dry and preaction systems.

Category	Included	Description
Steady state piping network analysis	Included	Friction loss, pump, Water Supply, discharge device, valve·PRV·orifice
Supply·Design Capacity·Combined Analysis	Included	Current-condition calculation, required-pump sizing by target pressure, and multi-pump roles
Fitting Equivalent Length	Included	Convert straight pipe length for losses such as elbows, tees, valves, etc.
Result/Warning Interpretation	Included	Node/Pipe/Pump Results, Critical Node, Alert Category
Dedicated Antifreeze Analysis	Included	Activated only in dedicated menu. Downstream D-W·Antifreeze Physical properties, water side sequential combination, clear parallel main pump support
Pipe Sizing	Included	Manual (head count) and automatic (velocity/hydraulic). A grid is sized from the flow in the design area that is currently active, and branch lines and cross mains are each unified to the largest diameter in the group.
Pipe geometry editing	Included	45° diagonal drawing; rotation of pipe runs, risers, and tees. Elevations and bends changed by rotation enter the calculation.
Water Delivery Time (WDT)	Reference	Covered separately in Engineering Overview
Water hammer/pressure wave	Excluded	Water hammer and elastic pressure waves are not analyzed directly
Replacement for Regulatory	Excluded	It is an auxiliary material for

Determination		design/review and does not replace licensing/supervision judgment.

0.2 Related standards and engineering models

Category	Reference	Role in this document
Compliance/reference standards	NFPA 13 Equivalent Length·Provisions related to internal diameter compensation	Reference concept for Equivalent Length of fittings and valves
Pipe Specifications	KS etc. Piping Actual Inside Diameter Material	Basis of input Nominal Diameter (DN) and Actual Inside Diameter
External Engine	EPANET / WNTR	Nonlinear piping network steady-state flow/pressure analysis
Engineering Theory	Continuity/energy equation, H-W, D-W, emitter	Friction·discharge·Pump operating point physics basis

The part expressed as “NFPA 13 standard” in this document refers to published design practices such as Equivalent Length·internal diameter correction. EPANET/WNTR hydraulic analysis and discharge·Pump modeling is not an NFPA provision itself, but an engineering model that K-Fire uses to calculate the piping network operating point.

0.3 Key terms

Term	Meaning
Steady State	Flow and pressure distribution at a stable operating point that does not change with time
H-W	Hazen–Williams. C-Factor based empirical friction loss equation
D-W	Darcy–Weisbach. Friction-loss equation based on friction factor f and absolute roughness ε
Equivalent length L_E	Fitting and valve losses converted to an equivalent straight-pipe length of the same diameter
K-Factor	Sprinkler/nozzle discharge coefficient. Q = K√P
Critical Node	Active discharge device with the smallest margin above the required pressure
Supply	Calculate flow rate and pressure based on given pump·Water Supply conditions
Design Capacity	Calculate the required pump conditions to satisfy the target minimum pressure
Combined Analysis	Simultaneous review of existing Supply pumps and new (calculated) pumps in one network
Antifreeze Boundary	Explicit node where the downstream Antifreeze section begins. Could be the main pump itself
Water side sequential combination	After obtaining the antifreeze-side Q–P boundary, determine pump pressure along the water-side

	path
Pipe Sizing	DN proposal in the design phase. Preview and apply manually (number of heads) or automatically (flow velocity·hydropower)

0.4 unit system

User input and reports use units familiar from fire design practice (bar, L/min, mm, etc.), and the calculation engine converts them to SI-based internal units.

Conversion	Relationship
Pressure ↔ Head	$H[m] = P[\text{bar}] \times 10.197$
Flow Rate	L/min ↔ m³/s (×1/60000)
Length/inner diameter	mm ↔ m

The pressure on the screen/report is generally based on gauge pressure [bar]. If necessary, indicate pressure and head together.

1. Overview

1.1 Purpose of Hydraulic Calculation

steady-state hydraulic calculation answers the following questions:

- What flow rate and pressure are reached at each discharge device under given pump·Water Supply conditions?
- What pump head and flow rate are needed to meet the target minimum water discharge pressure?
- In a configuration where parallel/series or existing/new pumps are mixed, how are the roles and operating points of each pump distributed?
- How are pipe diameter, friction coefficient, fittings Equivalent Length, valve condition, PRV·orifice reflected in the results?

The calculation assumes a steady state. Time-dependent pressure waves, water hammer, valve opening/closing transient response, air entrainment, and water delivery delay are not within the scope of this function. Water Delivery Time (WDT) refers to the WDT document.

1.2 Analysis flow overview

K-Fire converts the user-entered piping network into a hydraulic model that EPANET/WNTR can process, solves for the steady-state condition, and organizes the results by design node and pipe.

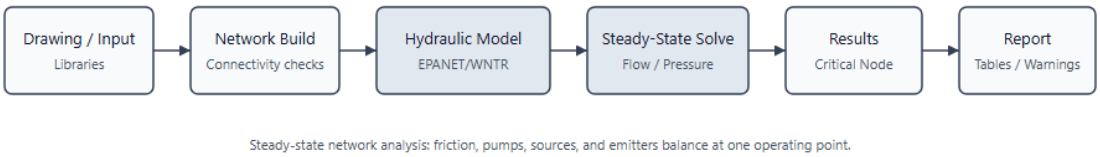


Figure 1-1. Hydraulic calculation analysis flow

Stage	Description
Drawing/Input	Piping, device, library, design settings
Network Construction	Network configuration for connectivity/property inspection and analysis
Steady state analysis	Operating point where friction, pump, Water Supply·discharge are balanced
Summary of results	Node·Pipe·Pump result, Critical Node, warning
report	Output for tables, summaries, and reviews

1.3 Supported analysis types

Mode	Purpose	Key Question
Supply	Flow rate/pressure of current pump·Water Supply conditions	“What flow and pressure result under the current conditions?”
Design Capacity	Required pump conditions to satisfy target minimum pressure	“What pump capacity is required?”
Combined Analysis	Review of existing and new pumps in one network	“How much is supplied by the existing pump, and how much is required from the new pump?”

The three modes share the same piping network and the same steady-state engine. What changes is the input conditions and goal settings.

1.4 Friction formula selection (H-W vs D-W)

Item	Hazen–Williams	Darcy–Weisbach
Typical Use	Firefighting piping network design practice, room temperature water	Close examination, check for roughness/Reynolds effects
Input Parameter	C-Factor	Absolute roughness ε [mm]
Flow Regime	Empirical expression, do not use Reynolds number directly	Calculation of laminar, transitional and turbulent friction coefficients
K-Fire Default	H-W	User selectable

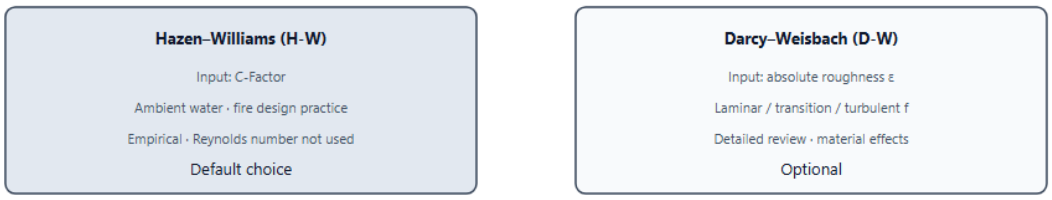


Figure 1-2. H-W and D-W selection concept

Even for the same piping network, H-W and D-W results need not match exactly. In the standard preset, C=120 and ε=0.100 mm are provided as corresponding values for wet steel pipe, but the two formulas differ in physical model and exponent structure.

1.5 Compliance/reference principles

1. The friction loss formula supports both H-W and D-W, and the headloss option of EPANET/WNTR is used in pipe network analysis.
2. Equivalent Length values for fittings and valves are managed separately for NFPA-based reference tables and user-adjusted values.
3. The pipe dimension used directly in the calculation is Actual Inside Diameter, not Nominal Diameter (DN).
4. Library default values are intended to aid input and should be checked against manufacturer-approved books, field specifications, and application standards.

2. Engineering basics

2.1 Fluids and Assumptions

The core hydraulic calculation is a steady-state piping-network analysis for water. Fluid properties are based on ambient-temperature water. Darcy–Weisbach auxiliary calculations use kinematic viscosity v = 1.004×10^{-6} m²/s and gravitational acceleration g = 9.80665 m/s².

Item	Assumption	Design Significance
Fluid	water, incompressible	Density changes are ignored in general water-based fire extinguishing equipment.
Flow	Steady State	Water hammer·filling propagation·transient response excluded
Temperature	room temperature properties	Special conditions with large viscosity changes are reviewed separately.
Pipe	Circular pipe, full pipe flow	Partially filled and open-channel flow are outside the scope
Loss	Pipe friction + fittings/valve loss	Straight pipe length and Equivalent Length or local loss
Discharge	Orifice/sprinkler type	discharge flow is proportional to the square root of pressure

The steady state result means a solution in which each node pressure, each pipe flow rate, and each pump head are balanced at the stable operating point at a specific point in time.

2.2 Continuity Equation

In incompressible steady flow, the incoming and outgoing flow rates at any junction must be preserved.

$$\sum Q_{\text{in}} - \sum Q_{\text{out}} - Q_{\text{demand}} = 0 (2-1)$$

Item	Meaning
Q_{in}	Piping flow into the node
Q_{out}	Pipe flow rate leaving a node
Q_{demand}	Head, nozzle, hose stream, etc. External discharge flow

The average velocity across the pipe cross-section is as follows.

$$V = \frac{Q}{A} = \frac{4Q}{\pi D^2} (2-2)$$

Velocity is an intermediate result used in friction-loss and discharge reviews; it also helps identify excessive velocity, unrealistic pipe diameters, and input errors.

2.3 Energy equation

The steady-state energy relationship between two points in a pipe network is expressed in the form:

$$\frac{P_1}{\gamma} + z_1 + \frac{V_1^2}{2g} + H_p = \frac{P_2}{\gamma} + z_2 + \frac{V_2^2}{2g} + h_f + h_m + H_v (2-3)$$

Symbol	Meaning
P/γ	pressure head
z	elevation head(elevation)
$V^2/2g$	Velocity Head
H_p	Pump head
h_f	Straight friction loss
h_m	fittings·Local losses such as valves and orifices
H_v	head dissipated in pressure reducing valves, etc.

The EPANET series solver solves a nonlinear system of equations that simultaneously satisfies the headloss relationship of each link and the continuity equation of each node. In firefighting piping networks, friction loss and discharge device pressure are often more direct in design decisions than velocity head itself.

2.4 Hazen–Williams formula

The Hazen–Williams (H-W) formula is a method to empirically calculate the steady-state piping friction loss of room temperature water. Because it does not directly require Reynolds number and absolute roughness, it is widely used in wet sprinkler piping network design.

The H-W head loss form widely used in the literature is as follows.

$$h_f = 10.67 \cdot \frac{L \cdot Q^{1.852}}{C^{1.852} \cdot d^{4.871}} (2-4)$$

Symbol	Meaning	SI units (Equation 2-4)
h_f	Friction head Loss	m
L	piping length	m
Q	volume flow rate	m³/s
d	inner diameter	m
C	Hazen–Williams coefficient	—

The form converted into practical units (flow rate L/min, inner diameter mm, length m, loss bar) also maintains the same index. If you select H-W, the C-Factor of each pipe becomes the key input for friction characteristics. The larger C, the smaller the friction loss.

For wet steel pipe, C≈120 is commonly used in H-W, while ε≈0.100 mm is used as the corresponding D-W value. Because H-W C and D-W ε are physically different parameters, the two modes do not always agree.

2.5 Darcy–Weisbach formula

The Darcy–Weisbach (D-W) equation computes friction loss by obtaining the friction factor f as a function of Reynolds number and relative roughness.

$$h_f = f \cdot \frac{L}{d} \cdot \frac{V^2}{2g} (2-5)$$

$$Re = \frac{V \cdot d}{\nu} (2-6)$$

In the turbulent flow region, standard engineering equations such as Swamee–Jain approximation are used.

$$f = \frac{0.25}{\left[\log_{10}\left(\frac{\varepsilon/d}{3.7} + \frac{5.74}{Re^{0.9}}\right)\right]^2} (2-7)$$

In laminar flow, f = 64/Re; in the transition region, laminar and turbulent values are interpolated. Absolute roughness ε is entered in mm and converted to engine units. D-W explicitly reflects viscosity and roughness, which is advantageous for detailed review; the current version fixes kinematic viscosity to 20°C water.

2.6 Local loss

Local loss is energy loss that occurs at a specific location, such as a valve, orifice, or rapid expansion/contraction.

$$h_m = K_m \frac{V^2}{2g} (2-8)$$

Symbol	Meaning
h_m	Local loss head
K_m	Local loss coefficient
V	Average velocity through the element

K-Fire deals with local losses in two ways.

5. Equivalent Length Method: Convert losses from elbows, tees, valves, etc. into additional length of straight pipe
6. Coefficient loss: Expressing pressure drop as a loss coefficient in orifices, etc.

Here K_m is a local loss coefficient, a different physical quantity from the sprinkler/nozzle K-Factor.

2.7 discharge Characteristics (Emitter)

The discharge flow of sprinkler heads and nozzles is proportional to the square root of pressure.

$$Q = K\sqrt{P} (2-9)$$

Symbol	Meaning	representative unit

Q	discharge flow	L/min
K	Head/Nozzle K-Factor	L/min/√bar
P	Discharge-device pressure	bar

EPANET emitters use a head-based $q = C_e\sqrt{H}$ form, so K-Fire converts practical-unit K-Factors into the engine internal coefficient.

Hosestream is not a pressure-dependent emitter, but rather a specified flow rate. In other words, it is a "condition for extracting the required flow rate to the outside," and is distinguished from devices where the flow rate automatically changes depending on pressure, such as heads and nozzles.

3. Network and components

3.1 Network Representation

The hydraulic calculation applies to a piping network composed of nodes and pipes. Nodes represent junctions and device locations, while pipes have length, inside diameter, and friction properties. Elevation is included in pressure calculations as elevation head.

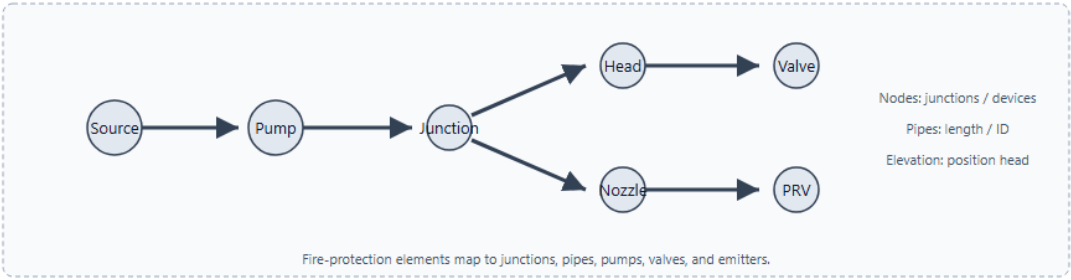


Figure 2-1. Network element concept

element	role
general connection point	Flow confluence/branching
Head/Nozzle	Pressure-dependent discharge
hose stream	Specified flow rate discharge
Pump	Adds Head
Water tank·Water Supply	energy reference point
Valve/check valve	Open/close/backflow restriction·Equivalent Length
PRV	Limit downstream pressure below set pressure
Orifice	local pressure drop

3.2 Piping properties

Item	Meaning	Calculation Treatment
Actual length L	straight pipe length	friction loss
Equivalent length L_E	Equivalent length of fittings and valves	Added to effective friction length
Inside diameter D	inner diameter	flow velocity·Friction loss
Nominal Diameter (DN)	Standard name	Equivalent Length Standard table matching
C-Factor	H-W friction coefficient	H-W mode
Absolute roughness ε	D-W roughness	D-W mode
Closed/Check	Valve/piping status	Path and backflow restrictions

The value directly used in the calculation is Actual Inside Diameter, not Nominal Diameter (DN). If the pipe specifications are changed, the friction loss and Equivalent Length correction may change together.

3.3 Connectivity and elevation

discharge device that cannot be reached from Water Supply, unconnected pipes, and isolated sections behind closed valves reduce the reliability of results and are displayed as warnings or errors.

Elevation differences are reflected in hydrostatic pressure terms. The height difference between the elevated water tank, basement, riser pipe, pump room, and protected area must be reflected in the elevation input. Even if the pipes are drawn on the same plane, the pressure varies if the elevation is different.

3.4 Hydraulic Significance of Components

Firefighting equipment elements are mapped to basic hydraulic element understood by the calculation engine.

design elements	Hydraulic Significance
Head/Nozzle	Pressure dependent discharge device
hose stream	Specified-flow demand
Tank/reservoir	Fixed head border
Water Supply type pump	Suction side Water Supply and pump curve
inline pump	Adds head within the piping
PRV	Pressure reduction based on set pressure
Relief Valve	Holds inlet set pressure, releases the excess (PSV)
Flow Control Valve	Limits flow to the set flow (FCV)
Orifice	Pressure drop due to local loss
inactive device	Removes discharge and pressure-reduction effects; connectivity may be retained

Only active heads/nozzles consume water. Inactive devices are not counted discharge.

3.5 Pipe Geometry Editing

From v6.0, pipes can be drawn and edited not only along the X, Y, and Z axes but also on 45° plan diagonals and in tilted directions. Because geometry edits carry straight into elevation and fitting detection, this section also explains how they enter the calculation.

Function	Description
45° diagonal drawing	Draws pipes at exactly 45° in plan. Length is entered as the actual pipe length.
Pipe run rotation	Rotates, as a rigid body, the whole pipe run that continues to one side of a pivot node — horizontally, vertically, or about its own axis. The angle is typed in or stepped in 1° increments, checked in a dashed preview, then applied.
Riser and tee rotation	A riser is tilted in two directions, toward X and toward Y. At a perpendicular tee, the branch or the line is turned about the opposite pipe, and the right angle is kept.
Bend limits	The actual bend at a rotated joint is limited to 90°. New drawing allows elbows up to 91°, and at a node that already has two or more pipes, only parallel or perpendicular pipes can be added.
Undo rotation	Returns exactly to the original coordinates and elevations, even after saving and reopening. If later edits change the shape of the run, the rotation is confirmed automatically.
Undo	Up to 30 steps can be undone.

Heights changed by rotation are written to node elevation, so the hydraulic calculation, the report, and Water Delivery Time all use the same values. Rotation is rigid, so pipe lengths and diameters stay the same, and heads on the rotated run tilt with the pipe (for example, water spray along an inclined conveyor). Fittings are always re-detected from the actual bend angle of the current geometry (§5.3).

A rotation that would overlap another pipe in 3D space, or a pivot node that cannot be rotated about (such as a cross or a non-perpendicular tee), is rejected before it is applied, with the reason shown.

3.6 Relief Valve

A relief valve opens when the pressure at its inlet exceeds the set pressure and releases the excess water to the outlet side, holding the inlet pressure at the set value. The calculation engine models it as an EPANET pressure sustaining valve (PSV). It is drawn as an in-line device with one inlet and one outlet, and the outlet pipe runs to a tank drawn by the user (a receiving tank or the pump suction tank).

Item	Description
Input	Set pressure (inlet, bar) — 12 bar when newly placed; loss coefficient — default 5.0; active state. The valve size is taken as the inlet pipe size.
Supply	When the inlet pressure reaches the set pressure, the valve opens and releases the excess water to the outlet; below it, the valve stays closed. The set pressure is gauge pressure at the valve elevation and, as with the PRV, is passed to the engine as head (§4.2).
Design Capacity·Combined Analysis	In a calculation with a pump being sized, the valve is treated as not operating, that is, as a closed valve, because a relief valve is normally closed at the design point.
Pipe Sizing	Treated as a closed valve while sizing runs.
Switched off (inactive)	Calculated as a closed valve. No water passes in either direction.
Flow direction	Water flows from inlet to outlet only. Pump suction-side detection and Water Delivery Time supply-source detection follow this direction, so a receiving tank beyond the relief outlet is not counted as a supply source, and an outlet returned to the pump suction tank is still recognized as part of the suction side.
Installation rule	[inlet pipe → relief valve → outlet pipe] is required. The pipes on either side may differ in size and may bend (angle type). A branch (tee) at the valve is not allowed.

Capacity shortfall warning	If, in the Supply calculation, the relief valve is releasing water but the inlet pressure stays above the set pressure, the valve is judged too small even when fully open, and a warning is issued. In that case, increase the valve size (the inlet pipe size) by one step.

Water released by the relief valve is included in the pump flow. The released amount is read as the flow in the pipe from the outlet to the receiving tank (pipe results table in the report). A tank is assumed by default to have backflow prevention at its outlet, so this setting must be switched off on a receiving tank for water to enter it. Even when the valve sits at a bend, no elbow equivalent length is added; the valve loss is represented by its loss coefficient.

Because the calculation is steady-state, the gradual opening, hysteresis, and chatter of a real valve are not represented. A recirculation layout that connects the outlet straight into the pump suction pipe, without a tank, cannot be separated from the suction side, so the Design Capacity calculation is blocked.

3.7 Flow Control Valve

A flow control valve limits the flow through the valve to no more than the set flow. The calculation engine models it as an EPANET flow control valve (FCV). It is installed as an in-line device, in the same way as a pressure-reducing valve (PRV).

Item	Description
Input	Set flow (L/min) — 1,000 L/min when newly placed; loss coefficient — default 5.0; active state. The valve size is taken as the inlet pipe size.
Supply	When the valve flow would exceed the set flow, it is held at the set flow. Where the system delivers less than the setting, the valve acts as an open valve with only its loss coefficient and does not increase the flow.
Design Capacity·Combined Analysis	In a calculation with a pump being sized, the valve is treated as not operating, that is, as a fully open valve with only its loss coefficient.
Pipe Sizing	Treated as a fully open valve while sizing runs, and the pipes on either side of the valve are proposed at the same diameter.
Switched off (inactive)	Calculated as a fully open valve with only its loss coefficient. Even when switched off, water does not flow backward through it.

Flow direction	Water flows from inlet to outlet only. Pump suction-side detection and Water Delivery Time supply-source detection do not pass backward through a flow control valve.
Installation rule	Same as the pressure-reducing valve. [inlet pipe → flow control valve → outlet pipe] is required, the pipes on either side must be the same size and in a straight line, and a branch (tee) at the valve is not allowed.

Pressure-reducing valves, relief valves, flow control valves, and orifices are handled together as special valves. An active pressure-reducing valve, a relief valve, and a flow control valve are one-direction devices whose incoming pipe is the inlet and outgoing pipe is the outlet; like check valves, they are judged by flow direction. Automatic equivalent-length detection does not substitute a library fitting for a special valve; the valve loss is represented by the valve loss coefficient.

A switched-off special valve is drawn dimmed in light gray with a dashed outline on screen and in report drawings, to distinguish it from an active valve. In the report, the NFPA calculation sheet breaks at a special valve, and the section upstream of it starts on a new sheet.

4. Pump·Water Supply·Boundary

4.1 Pump performance curve

A pump is a device whose output varies depending on the flow rate. K-Fire can create a standardized curve based on rated flow rate and rated pressure, or use a multi-point curve entered by the user.

$$H_p = F(Q)(4-1)$$

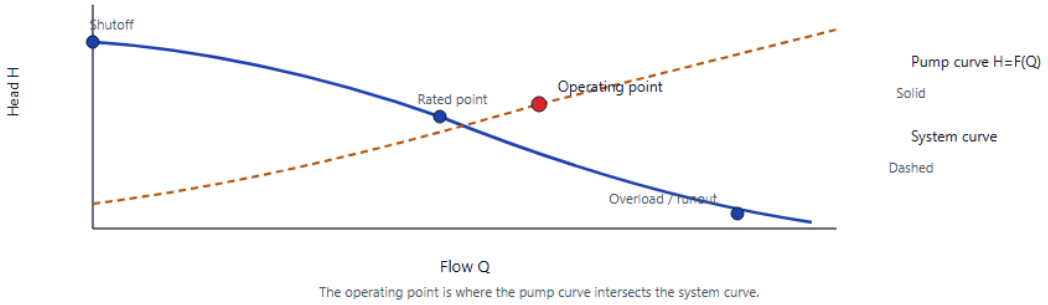


Figure 4-1. Pump curve and operating point

Autocurves generally include the following concepts:

Point	Meaning
Shutoff Point	head at discharge flow rate 0
Rated Point	Rated operating point entered by user
Overload Point	head at a flow rate greater than the rated flow rate
Runout	High flow endpoint where head is close to 0

The user-defined curve requires the flow rate to be monotonically increasing and head to be monotonically non-increasing. The pump operating point is the point where the pump curve and the system curve meet. After calculation, the flow rate, suction pressure, discharge pressure, and head of each pump are provided as results.

AutoCurve is a standardized approximate model for early design review. The manufacturer's allowable range must ultimately be checked with the manufacturer's pump curve.

4.2 Water Supply·Boundary conditions

Pipe network analysis requires at least one energy reference point.

element	role
Water tank (WT)/reservoir	Fixed head based on elevation plus water level
Water Supply type pump	Supply from the suction-side Water Supply and pump curve
inline pump	head increase between upstream and downstream

Design Capacity Target pump	Search for the required head and flow to meet the target pressure

The water tank standardhead is interpreted as follows.

$$H_{\text{source}} = z_{\text{node}} + h_{\text{waterlevel}} (4-2)$$

The set pressure of the PRV is delivered to the engine in head units.

$$H_{\text{set}} = P_{\text{set,bar}} \times 10.197 (4-3)$$

A model without an active Water Supply becomes a system of equations without a physical reference. Report review must verify actual Water Supply, pump suction conditions, and water tank level.

5. Equivalent Length and Fittings

5.1 Effective friction length

Equivalent length converts fitting and valve losses into an equivalent length of straight pipe of the same diameter. The effective friction length is formed by adding the sum of equivalent lengths to the actual length L.

$$L_{\text{eff}} = L + \sum L_E (5-1)$$

5.2 Inner diameter/C-Factor correction

The standard table Equivalent Length assumes the standard internal diameter and standard C. If the actual pipe inner diameter and C-Factor are different, the following type of correction is applied.

$$L_{E,D} = L_{E,0} \left(\frac{D_{\text{actual}}}{D_{\text{ref}}} \right)^{4.87} (5-2)$$

$$L_E = L_{E,D} \left(\frac{C}{120} \right)^{1.85} (5-3)$$

Item	Meaning
$L_{E,0}$	Standard table Equivalent Length
D_{ref}	Standard inner diameter assumed by the standard table
D_{actual}	Actual inner diameter of project pipe
C	Hazen–Williams coefficient for the pipe in question

Piping with a larger C (smoother) requires a longer length of straight pipe to produce the same loss, so Equivalent Length increases.

5.3 Automatic fittings Judgment Concept

After estimating the flow direction, the inflow/outflow relationship at each node is analyzed.

Classification	Concept
elbow	Change in direction of pipe bending
tee branch pipe	Direction of side pipe at branch/join
Straight-through Tee	Direction straight ahead. It is common to treat losses as 0
Valve	By valve type Equivalent Length

If there is no library cell or the DN/inner diameter is outside the standard range, do not make unreasonable estimates. The item is marked as a warning, allowing the user to correct the data.

fittings Equivalent Length can vary depending on the flow direction, so the concept of repeating analysis and fittings recalculation until stable is used. This is different from simple serial addition, which is “just add once and you’re done.”

5.4 Source classification

Fitting-library values are managed by source.

Category	Meaning
Standard table raw value	Values from NFPA series standards
user edit	Modify the standard value based on project judgment
user input	Directly enter values that are not in the standard table

Technical reviewers must be able to track which values come from the benchmark and which are project calibrations.

6. Analysis mode

6.1 Mode Overview

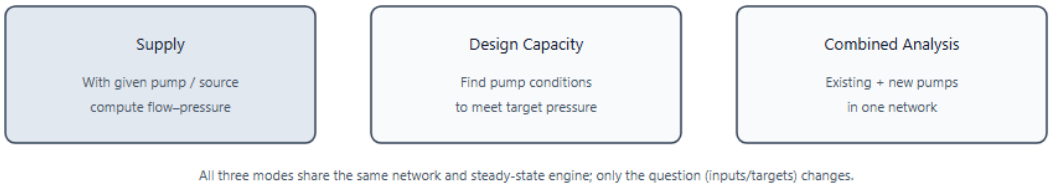


Figure 6-1. Analysis mode concept

The three modes ultimately share EPANET/WNTR steady-state analysis and result reconstruction.

6.2 Supply

Supply calculates the flow rate and pressure using the pump, Water Supply, valve, piping, and discharge device conditions entered into the current project. Since fittings Equivalent Length depends on the flow direction, fittings reflection is repeated until the direction is stable, and then the final solution is confirmed.

6.3 Design Capacity

Design Capacity finds the necessary pump conditions so that the active discharge device satisfies the required minimum pressure. The basic target pressure is the minimum required pressure of the design setting, and if there is a separate required pressure for each head, that value may take priority.

Design Capacity is different from simple serial calculations that “calculate pump pressure by adding friction loss once.” discharge flow changes depending on pressure, and in a piping network with intertwined parallel paths, nonlinear analysis must be repeated to obtain results close to the actual operating point.

6.4 Combined Analysis

Combined Analysis analyzes some pumps as existing Supply pumps and some pumps as new or calculation target pumps simultaneously within one piping network. The existing pump maintains the input curve and obtains the necessary conditions so that only the pump subject to calculation satisfies the target pressure.

It can be used to calculate new pump specifications when expanding existing facilities, reviewing parallel operation of pumps, and expanding areas.

6.5 Critical Node

The targets of pressure review are pressure target nodes such as active heads and nozzles. Calculate the difference between the current pressure and the required pressure for each target, and select the node with the smallest margin as Critical Node.

$$\text{margin}_i = P_i - P_{\text{req},i} (6-1)$$

Critical Node is the point that designers should review first. Unreachable nodes or inactive devices may be excluded from Water Supply, so you should check the result table and warnings together.

6.6 Dedicated Antifreeze Analysis

K-Fire is a water-first application. Floating projects and the general Run Analysis workflow handle water only; an Antifreeze analysis is explicitly created and run only through the dedicated workflow in the Tools menu. Design settings, a change of friction formula, or pump selection alone do not enable Antifreeze mode.

6.6.1 Scope of application

Item	Description
Scope	Steady-state analysis of a fire-protection Antifreeze system
Fluid	Incompressible/Newtonian fluid. Density and viscosity snapshot for one point of design temperature
Antifreeze Section	Antifreeze material properties only downstream of the border node that the user has picked in the drawing. The main pump can be placed as a boundary (virtual Water Supply)
friction formula	Antifreeze section is fixed to Darcy–Weisbach
fittings Loss	NFPA Equivalent Length Basic. Block calculation when required data is missing
Excluded	FDT/WDT, general multi-fluid, Antifreeze pump viscosity correction, multi-Antifreeze zone simultaneous analysis, series/unclear pump topology
water side pump	A single main pump or a distinct group of parallel pumps identified at the boundary. Suitability determination based on actual flow rate and individual curve for each pump

6.6.2 Boundary and Sequential Combinations

A typical topology is as follows:

Main pump (virtual Water Supply·boundary possible) → [Water side] → Boundary → [Antifreeze piping] → Head·Nozzle

The calculation order does not mix water and Antifreeze in a single EPANET model.

7. Analyze the downstream Antifreeze section using D-W and the Antifreeze density and viscosity to determine the required boundary flow and pressure.
8. If the boundary is the main pump, the required pump pressure is equal to the boundary required pressure.
9. If the boundary is downstream of the pump, the pump operating point is obtained by analyzing the pump → boundary path with fixed demand on the water side (project H-W or D-W·water properties).

10. If there is a project pump H–Q curve, the available pressure corresponding to the pump flow rate is read from the water reference curve to determine the residual pressure. Antifreeze Viscosity correction is not applied to the pump.

The water side supports only a single main pump or a clearly identified parallel pump group at the boundary. A parallel group is analyzed as a common-head manifold, and each individual curve is evaluated using that pump's actual flow. Curves are not superimposed into a synthetic pump curve, and the group is acceptable only when every participating pump is acceptable. Series pumps, direct pump-to-pump connections, auxiliary pumps outside the calculated flow path, and configurations with missing or ambiguous analysis flow fail closed. An Antifreeze fill pump is not included in the fire-flow solution.

6.6.3 Fluid Properties, Discharge, and Compatibility

Calculation inputs are constant property snapshots stored in the project (product name, design temperature, density, either dynamic or kinematic viscosity, source). Even if the catalog changes later, the existing case will not be quietly changed. If density or viscosity are missing, calculation is blocked without replacing with water.

Heads and nozzles are solved as emitters by converting the water-based K for the fluid density. If an active discharge device has no K and the library cannot resolve one, the program uses the default K=80 and records a warning and the node ID in the results and PDF.

Antifreeze Projects containing analysis cases are stored separately so that older versions of apps do not quietly discard the cases and refuse to open them. Water-only projects will maintain their existing water-only format. If you delete a case, the drawing topology remains the same and returns to the water-only state.

6.6.4 Interpretation of results

The primary value shown on the dedicated results screen and PDF is the required pump operating point (flow and pressure). The required boundary point and water-side pressure difference are supplementary information. The report preserves a snapshot of the water-side and Antifreeze-side flow paths, water-side node pressures, pump discharge pressure, check-valve upstream direction, actual flow by pump, and each individual curve evaluation at calculation time. In addition to the common hydraulic results table, the Antifreeze section presents head loss, Reynolds number, flow regime, and friction factor in a detailed D-W table; pump charts are drawn separately for each pump. Warning banners may identify missing fitting data, non-turbulent D-W flow paths, unsuitable or unevaluated pump curves, and fallback to the default K=80.

6.7 Pipe Sizing

K-Fire provides a Pipe Sizing function that suggests piping Nominal Diameter (DN) at the design stage. The same drawing, library, and design settings as general hydraulic calculation are used, and the user confirms application after checking the change plan in preview. The calculation result is a design aid, and the final pipe diameter is determined by application criteria and technician review.

6.7.1 Calculation method

11. Manual (head count): proposes branch and main diameters from the DN table in the pipe-sizing library, using the number of downstream heads. Suited to tree networks. Head-count sizing is not applied when the network contains a grid whose branch lines can receive water from both ends.
12. Automatic (velocity/hydraulic basis): Run Supply (Demand) analysis to obtain flow by pipe, then propose pipe diameters from the allowable velocity and DN ladder. Repeat analysis → proposal → reanalysis until DN values stabilize. If the process does not converge, adopt a conservative DN envelope and issue a warning.

6.7.2 Uniform diameter on a grid

Automatic sizing uses only the Supply calculation for the design area that is currently active. It does not move that area and calculate again. Whether another location is more demanding is checked by the user, by moving the area and running Supply.

On a grid, after diameters are proposed from that flow, branch lines are set to one diameter and cross mains are set to one diameter. Each group takes the largest diameter calculated within the group. A branch line is a run that carries heads. A cross main is a horizontal run with no heads.

A network is treated as a grid when a branch line that carries heads can receive water from both ends. A tree branch fed from one end only, and a loop whose closed main carries only tree branches, are not unified this way; they follow the ordinary automatic velocity path in §6.7.1. Upright-and-pendent heads are not treated as a grid, because that arrangement cannot be distinguished from a two-head tree branch.

6.7.3 Precautions when applying

- Preview and application are separate, and the drawing DN does not change before application.
- pipe diameter specified as a manual exception can be maintained, but exceeding the limit flow velocity is displayed as a warning.
- If there is no active discharge device, no automatic calculation is performed (preventing the entire interval from being reduced to at least pipe diameter).
- After calculation, be sure to recheck the pressure·flow velocity·pump operating point using Supply·Design Capacity, etc.

7. Input data

hydraulic calculation Accuracy is directly dependent on drawing input, libraries, design settings, and pump·discharge device specifications.

7.1 Design settings

Setting	Meaning
friction formula	H-W or D-W
minimum required pressure	Basic target pressure of Design Capacity·Critical Node

default roughness	Default absolute roughness in D-W mode

If there is a value specified for an individual pipe, it takes precedence over the global default value. In H-W, C-Factor is the key, and in D-W, absolute roughness is the key.

7.2 Drawing/device input

Input	Primary Effect
Pipe length·Actual Inside Diameter·DN	Friction loss, flow velocity, Equivalent Length matching
Elevation	Hydrostatic pressure·elevation head
Pump curve/rated point	Operating point, Design Capacity required specifications
Head/Nozzle K-Factor	discharge flow·Pressure distribution
Specified hose-stream flow	fixed demand
Valve opening/closing/check direction	Path/backflow
PRV set pressure	Downstream pressure upper limit
Relief valve set pressure	Inlet pressure upper limit, releases the excess
Flow control valve set flow	Valve flow upper limit
Orifice loss coefficient	local pressure drop

7.3 Libraries

Library	role
Pipe	By standard/DN Actual Inside Diameter
Fittings/Valves	Equivalent Length Reference values and sources
Head/Nozzle	K-Factor or reference Q–P
Pump	Rating/curve input assistance
Pipe Sizing	Number of heads → DN standard table (manual calculation). Standard management for each project

Calculating nozzles can inversely calculate the K value from the standard pressure and standard flow rate.

$$K = \frac{Q_{\mathrm{ref}}}{\sqrt{P_{\mathrm{ref}}}} (7-1)$$

Having library snapshots stored in a project ensures that the calculation basis for past projects does not suddenly change when common libraries are updated.

7.4 Input validation principles

Validation Item	Purpose
Connectivity	Is it possible to reach from Water Supply to discharge device?
Pipe Properties	Prevent omission of length, inner diameter, DN, and

	friction coefficient
Valve Status	Closed valves and isolated sections
Pump Curve	Monotonicity and rated point matching
Discharge Device	Active state, K value, required pressure
Library Reference	Detection of deleted/changed items

Verification does not replace designer judgment. It serves to raise warnings about dangerous omissions and to clarify which inputs the results depended on.

8. Output and result interpretation

Results are provided based on the nodes and pipes entered by the user. There is no need to directly interpret the auxiliary expressions inside the calculation engine.

8.1 Summary and Critical Node

Item	Meaning
Total discharge flow	Summed flow rate of active discharge device
Calculation Method	H-W or D-W
Critical Node	Minimum margin compared to required pressure
Critical Pressure	Pressure of Critical Node
discharge device number active	Size to be reviewed
Pump Allocation	Each pump operating point/role

If the Critical Node pressure is at or above the required pressure, the reviewed nodes may be interpreted as satisfying the pressure requirement. However, warnings or abnormal values preclude determining compliance from the Critical Node alone.

8.2 Node/piping/pump results

Category	Primary Item
Node	Pressure [bar], head[m], discharge/demand
Before and after device	Upstream and downstream pressure of pump, PRV, orifice
Pipe	Flow, flow velocity, loss
Pump	Flow rate, suction pressure, discharge pressure, head, warning

The direction of pipe flow is analyzed along with the connection relationship. flow velocity is useful for finding excessive friction losses, unrealistic pipe diameter and input errors. Negative pressure and backflow on the suction side of the pump are reviewed as separate warnings.

8.3 Result interpretation flow

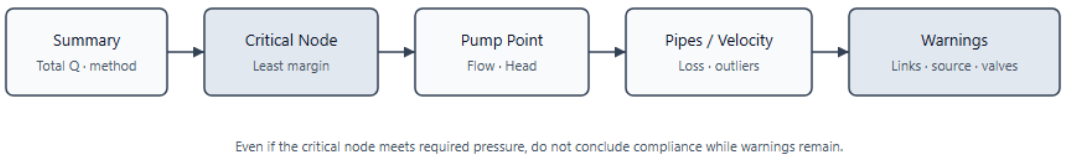


Figure 8-1. Results interpretation flow

Recommended reading order for the report:

13. Analysis mode, friction formula, active discharge device range
14. Critical Node and required pressure and margin
15. Pump operating point (Supply) or required specifications (Design Capacity·integrated)
16. Piping flow velocity·Outlier value of loss
17. Warning·Isolation·Water Supply Review conditions
18. If necessary, verify intermediate values with auxiliary calculations for a single pipe, nozzle, or similar component

8.4 Interpretation of warnings

A warning does not just mean a calculation failure. Even if the calculation has been completed, it represents conditions that the designer must review.

Category	Meaning
Solver warning	Analysis Imbalance, negative pressure, disconnection, etc.
pump warning	Backflow, suction side vacuum, abnormal operating point
Fitting Warning	DN/library cell missing, etc.
Device warning	Relief valve capacity shortfall (inlet stays above the set pressure even when fully open)
Topology warning	Isolated section, unreachable node
Result warning	Failure to organize some results/abnormal values

Results with warnings do not assume that Critical Node is suitable simply because it satisfies the required pressure.

8.5 Report Output

The report cover shows the whole piping network of the first scenario as an isometric drawing. Every pipe is drawn as a line, whether or not it carries flow, and heads and nozzles as dots, in the same view direction as the isometric drawings in the body. It is a vector drawing, not a screen capture, so it stays sharp in print.

Only scenarios with saved calculation results are included in the report. If any scenario has no results, the program lists those scenario numbers before output and asks whether to continue. To include a scenario, first use ‘Save Results’ in the Scenario Manager.

9. Assumptions/Limitations/Application Limits

Specifying limitations is a condition for safe engineering interpretation of results.

9.1 Limitations of steady-state assumption

The following phenomena are not subject to direct interpretation.

Excluded Phenomenon	Description
Water hammer/pressure wave	Elastic transient response due to rapid opening/closing/starting
Filling and air discharge	Dry side water reach/trip. WDT range
Partial filling/stratification	Two-phase flow details, not just a full tube
Temperature dependent viscosity	Correction of physical properties other than those fixed at room temperature. However, Antifreeze-specific analysis uses a 1-point snapshot of the design temperature.
Control logic details	Operation by detector, panel, and manufacturer

The result is an engineering estimate of the stable operating point created by the input piping network and device characteristics. If there is a request for field acceptance testing, manufacturer data, or AHJ, the relevant data takes priority.

9.2 Restrictions on input and use

Pipe Actual Inside Diameter, length, elevation, friction coefficient, Equivalent Length, head K-Factor, pump curve, Water Supply conditions all greatly affect the results. Library default values are intended to aid design input and should be compared with manufacturer-approved books, field specifications, and application standards.

K-Fire results are engineering calculation data to support design and review. Statutory licensing, supervisory approval, and final responsibility determination are subject to project application standards and technical review.

9.3 Relationship with WDT

steady-state hydraulic calculation deals with “the final operating point of a water-filled net.” WDT deals with “the process of filling up a network that is not yet full over time.” During WDT calculation, the water flow rate and pressure refer to the steady-state analysis, but filling the intermediate value of the progress state and the final operating point should not be directly compared one-to-one.

9.4 Limitations of Antifreeze Analysis

- The general Run Analysis and water-only results are numerically separated from the Antifreeze case.
- EPANET Do not include water and Antifreeze properties at the same time in one model. Combinations are sequential.
- Pump curve conformance is H–Q based on water, Antifreeze without viscosity correction.

- Simultaneous analysis of multiple Antifreeze zones and FDT/WDT Antifreeze are not supported. For water-side main pumps, only single or definite parallel groups are allowed, and series or unclear topologies are failed and closed.
- If manufacturer physical property/device data are not available, preliminary or warning status may be left.

9.5 Limitations of Pipe Sizing

- Pipe Sizing is a pipe diameter suggestion tool and does not replace the final conformance decision.
- Manual head-count sizing does not apply to a grid (a network whose branch lines can receive water from both ends).
- Automatic sizing depends on the flow of the heads that are active now, the velocity limit, and the DN ladder, so an input error distorts the proposed DN as well.

10. Verification approach

K-Fire's hydraulic calculation verification consists of the following categories:

Category	Validation Perspective
Unit/Physical formula	Pressure/flow unit conversion, H-W/D-W, emitter, Equivalent Length correction
Network	Connectivity, elevation, valve status, Water Supply boundary
Analysis Mode	Supply Operating point, Design Capacity Target pressure, Combined Analysis Pump distribution
Antifreeze	Downstream D-W, density correction emitter, water side sequential combination, flow rate/individual curve determination for each parallel pump (based on water)
Result/Warning	Critical Node, pump diagnostics, and fitting/topology warnings
Pipe Sizing	Head-count and velocity sizing, uniform branch-line and cross-main diameters on a grid, and DN consistency before and after the sizes are applied.
External Engine	Numerical accuracy and limits for EPANET/WNTR are within the scope of the respective engine documentation/licenses.

K-Fire provides firefighting equipment modeling, unit conversion, fittings Equivalent Length, analysis mode, and report layer on top of a third-party hydraulic analysis engine.

Appendix

Appendix A. Main symbols

Symbol	Meaning	representative unit
Q	volume flow rate	L/min, m³/s
P	Pressure	bar
,	Head, Pump Head, and Head Loss	m
,	Piping Actual Inside Diameter	mm, m
L	actual pipe length	m
L_E	Equivalent length L_E	m
L_{eff}	effective friction length	m
C	Hazen–Williams coefficient	—
ε	Absolute roughness ε	mm, m
f	Darcy friction coefficient	—
Re	Reynolds number	—
V	Average Velocity	m/s
K	Head/Nozzle K-Factor	L/min/√bar
K_m	Local loss coefficient	—
z	Elevation(elevation head)	m
g	acceleration of gravity	m/s²
ν	Kinematic viscosity coefficient	m²/s

Appendix B. List of Figures

Number	Title	File
Figure 1-1	Hydraulic calculation analysis flow	figures/hydraulic/fig_1_1_analysis_flow.svg
Figure 1-2	H-W and D-W selection concept	figures/hydraulic/fig_1_2_hw_dw_choice.svg
Figure 2-1	Network element concept	figures/hydraulic/fig_2_1_network_elements.svg
Figure 4-1	Pump curve and operating point	figures/hydraulic/fig_4_1_pump_curve.svg
Figure 6-1	Analysis mode concept	figures/hydraulic/fig_6_1_analysis_modes.svg
Figure 8-1	Results interpretation flow	figures/hydraulic/fig_8_1_result_flow.svg

Appendix C. References

The cited edition, section, and table numbers are confirmed by comparing them with the original text.

19. NFPA 13, Standard for the Installation of Sprinkler Systems.
20. EPANET hydraulic analysis documentation.
21. WNTR documentation and license notice.
22. Hazen–Williams and Darcy–Weisbach headloss formulations.
23. Swamee–Jain approximation for turbulent friction factor.

24. Orifice / sprinkler discharge relation .

— End of document (v1.5.0) —